

Humayan Kabir



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Profile

Machine learning and data engineer with three years across AI, production data infrastructure, backend services, and high-availability operations. Currently working on RAG-based retrieval and LLM-based document-generation systems for a legal practice. Recently built Mercato, a multimodal bilingual marketplace search engine demo with a learning-to-rank layer, deployed on Google Cloud Run.

Skills and Tools Used

- **Machine Learning:**
 - Python, PyTorch, Transformers
 - scikit-learn, CNN, GNN
 - RAG, LLM integration
 - bge-m3, OpenAI CLIP, LightGBM
- **Retrieval and Ranking:**
 - BM25, FAISS, HNSW, Optuna
 - Reciprocal Rank Fusion, LambdaMART
- **Data and Pipelines:**
 - Apache Spark, Apache Airflow
 - HDFS, Parquet
- **Backend and APIs:**
 - Django REST Framework, FastAPI
 - Go (in progress)
- **Database and SQL:**
 - PostgreSQL, TimescaleDB, MySQL
- **Cloud, Containers, and Operations:**
 - Google Cloud Platform (Cloud Run, Cloud Scheduler)
 - Hugging Face Spaces, Docker
 - Zabbix, Grafana, Unix Shell Scripting

Professional Experience

Reliance Legal Services – AI Solutions Engineer (Contractual)

Oct 2025 – Present · Dhaka, Bangladesh

- Architected and built a RAG-based retrieval pipeline over historical briefs, contracts, and memos, covering document ingestion, chunking, embedding, and query-time retrieval.
- Built an LLM-backed document-generation service, reducing per-case formatting time from 30 minutes to under 1 minute.

Banglalink Digital Communication · via Searchlite Communications · Oct 2022 – Jun 2024

Specialist Operations Support Systems Developer · Jan 2023 – Jun 2024 · Dhaka, Bangladesh

- Ran the high-availability monitoring, alerting, and visualization platform behind operations support systems.
- Rebuilt the platform on PostgreSQL and migrated traffic in phases with no downtime, reducing storage by about 50%, CPU usage by over 20%, and alert-processing overhead by about 15%.

Big Data Analyst · Oct 2022 – Dec 2022 · Dhaka, Bangladesh

- Built Spark and Airflow pipelines feeding KPI values to the API team for a new service launch.
- Evaluated distributed ETL tooling across HDFS, Spark, and Airflow.

adorsho praniSheba Ltd (Shurjomukhi Group) – Python DevOps Engineer

Jan 2022 – Sep 2022 · Dhaka, Bangladesh

- Built and shipped containerized Django REST applications with MySQL, deployed with Docker.

Independent Projects (Jul 2024 – Present)

Mercato: Marketplace Search Engine Demo ([Google Cloud Run](#), [Hugging Face Spaces](#), [GitHub](#)).

- Multimodal multilingual search engine with 3 retrieval channels (BM25, bge-m3, CLIP), merged by reciprocal rank fusion (RRF) and reordered by per-locale LambdaMART models.
- Built on the Amazon Shopping Queries (ESCI) dataset, serving English and Japanese locales.
- Adding the image channel to the two-way text fusion raised NDCG@10 from 0.4545 to 0.4821, the reranker added 0.0319. HNSW cut median query time from 132 ms to 0.31 ms.

The Round Wall: Neural Cryptanalysis of SHA-256 (Python, CNN, GNN)

- Measured how far neural networks can predict reduced-round SHA-256 output, finding the limit at the 4th of 64 repeating rounds before accuracy drops to chance.
- Reproduced a published Speck32/64 distinguisher (77% at 5 rounds) as a control.

Education

North South University, Dhaka, Bangladesh (Jan 2022)

- Bachelor's in Computer Science and Engineering, majoring in Machine Learning
- Thesis project: Real-time American and Bangla sign language detection (CNN, RNN, ResNet). Built the Bangla dataset and a live sign-to-text pipeline.

Blog

- Mercato: Marketplace Product Search
- The Round Wall: SHA-256 Cryptanalysis
- Kavel: Marketplace Listing Assistant
- humayankabir-re.github.io/blog

Language

- English – Fluent
- Bengali – Native